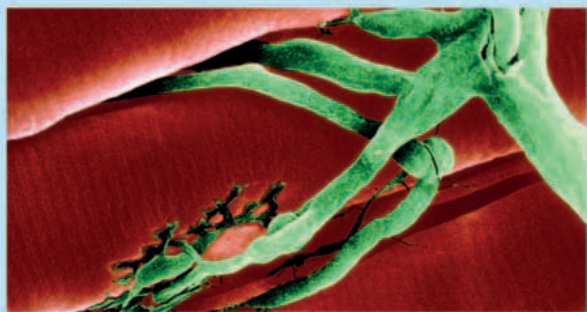


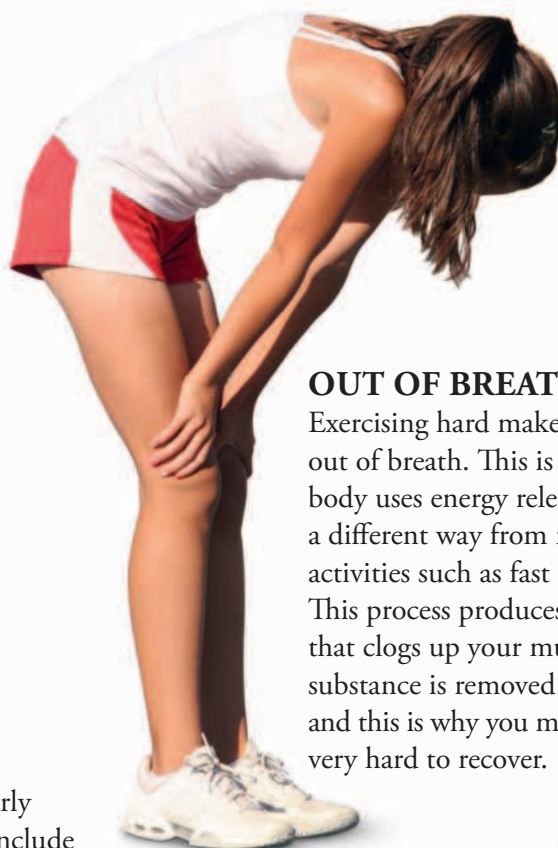


## LOOK CLOSER: GETTING THE MESSAGE

Your muscles are told to contract by nerve signals. These signals are carried from your brain by neurons (nerve cells) inside the nerves linked to your muscles.



▲ **NERVE-MUSCLE JUNCTION** *This is the place where a neuron (green) makes contact with a muscle fiber (red) of a skeletal muscle.*



## OUT OF BREATH

Exercising hard makes you feel out of breath. This is because your body uses energy released in a different way from normal for activities such as fast running. This process produces a substance that clogs up your muscles. This substance is removed by oxygen, and this is why you must breathe very hard to recover.

## MUSCLE FOOD

If you want strong muscles, you need to exercise regularly and eat foods rich in muscle-building proteins. These include foods such as meat, fish, and eggs. You also need other foods that contain carbohydrates, such as bread, potatoes, and rice. These supply your muscles with the energy needed to move you.

Foods rich  
in protein



WOW!

The fastest muscles in the body are the ones that move your eyeballs. They shift your gaze in about two-hundredths of a second.

Straightening  
your arm

*As your arm  
straightens, your  
biceps relaxes.*

*Your triceps pulls  
and gets shorter and  
thicker to make your  
arm straighten.*

*You were born with all your muscle fibers in place.*

